

Shreejal Bhattarai

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EDUCATION

B.S. Electrical Engineering, Comp. Engineering – Honors College

Texas State University, San Marcos, TX

GPA: 4.0/4.0

Expected Dec 2027

SKILLS

Programming/Automation: C/C++, Bash, Python, QCoDeS, LabView, Git, Linux/Unix, SystemVerilog

Lab Equipments: Oscilloscopes, Source Measurement Units (SMUs), Waveform Generators, Automated Test Equipment

Circuit Design and Simulation: LTSpice, KiCAD, MATLAB, Microwind, Altium, Analog Design

Data Analysis/Machine Learning: NumPy, Pandas, sci-kit learn, PyTorch

EXPERIENCE

Core Research Operations (CRO), Texas State University

San Marcos, TX

Student Engineer and Trainer

May 2026 - Present

- Conducted 100+ hours of operation, maintenance and training on an automated semiconductor wafer probing instrument for undergraduate and graduate coursework, supporting data collection and verification for multiple projects.
 - Author and revise Standard Operating Procedures for laboratory instruments including the *Cronus* automated Wafer Probing Station, *ezHEMS* Carrier Concentration Measurement, *PPMS*, and *VSM* improving consistency and safety compliance.
 - Train and certify researchers on equipment operation while tracking maintenance issues and managing equipment downtime to minimize disruptions.
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PROJECTS

Automated Test Fixture

July 2026 - Present

- Developing an automated test workflow for thin-film dielectric breakdown characterization by integrating a Hermes capacitance measurement system with a Keithley 2611B Source Measure Unit using GPIB to USB interface.
- Programmed the SMU to perform stepped-voltage stress tests while monitoring leakage current, automatically terminating the test at a predefined current threshold and acquiring stabilized measurements.
- Automated measurement sequencing, current averaging, parameter/metadata logging, and data export to improve measurement repeatability and reduce manual intervention.

Quantum Dot Device Fabrication and Characterization

March 2025 - Present

- Fabricated 20+ heterostructure devices by stacking with hexagonal boron nitride (hBN) and built an automated characterization pipeline using Python's QCoDeS library.
 - Validated device structures using the Atomic Force Microscopy (AFM) topographical mapping and characterized carrier transport using Hall-voltage measurements and Source Measure Units.
 - Fabricated multiple devices using thermal probe-based lithographic patterning.
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LEADERSHIP & INVOLVEMENT

Institute of Electrical and Electronics Engineers, Texas State Chapter

San Marcos, TX

Secretary

May 2025 – August 2026

- Secured \$10,000+ in funding by presenting strategic initiatives to IEEE Central Texas Section Board, supporting a 200+ member student organization.
- Directed a team of 30+ officers and interns to coordinate 25+ industry visits and speaker talks, contributing to IEEE's growth as the largest student organization in the College of Science and Engineering.

Student Learning Assistance Center, Texas State University

Aug 2024 – Present

Supplemental Instruction Leader

- Facilitated Learning Sessions for more than 1000+ students to date for differential equations, resulting an average improvement by a letter grade and a half.

Honors:

Scholarships: Texas State President's (Full) Scholarship, U.N. and Praja Pandey Endowed Award, Brian Wong Endowed Award, Allen and Sandra Kiesling STEM Scholarship, Quantum Rising Star Award, IEEE Career Catalyst Scholarship

Conferences: SIGHPC Immersion Program Awardee, Rice University Quantum Student Success Summit, Hot Chips Conference